Temporally Validated Forecast Combinations on Two Benchmark Collections

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Abstract

Forecast combinations estimate which past expert errors remain relevant at a new origin. We compare a fixed context-weighted rule with two variants whose parameters are selected using causally completed historical validation blocks. The original fifteen-case result remains unfavorable for fixed ContextMemory. In a separate external comparison, ValidationMix has mean series MASE 0.2855 versus RecencyMean's 0.3109 on 294 eligible M4 Weekly series, and 0.8993 versus 0.9085 on 1,422 unique M3 Monthly series. Exploratory paired-series bootstrap intervals for the respective differences are [-0.033914,-0.017227] and [-0.013176,-0.005173]. The monthly mean gain does not imply that most series improve: only 589 of 1,422 have strictly lower MASE than RecencyMean. AutoTheta retains lower monthly point error and lower interval score in both collections. All combinations share five local experts and at most twelve completed historical origins. Historical candidate evaluation excludes outcomes unavailable at its own query, followed by a final refit on completed outer-prefix memory. The regularized simplex solver enumerates every nonempty expert face; its mathematical uniqueness does not imply predictive superiority. The retained 53,148 full-precision method-origin paths support independent metric rescoring and reconstruction of the stored weighted mixtures. Expert training and external comparator fitting are separate operations. The uncertainty analysis preserves series as the resampling unit and is explicitly post hoc, unadjusted and conditional on the admitted collections. These bounded results support temporally validated combinations relative to their recency control; they do not establish priority for contextual weighting, an isolated causal benefit of its kernel, calibrated prediction intervals or population-wide superiority.

1 Introduction and research question

Combining forecasters can reduce dependence on one modelling assumption, but estimated expert weights can introduce another source of error. When a series changes regime, a recent expert loss may be more relevant than a long-run average. When an earlier regime recurs, a remembered context may instead carry useful information. The concrete question is whether a small context-distance weighting rule improves a combination beyond an otherwise matching recency-only estimate.

Feature-based forecast averaging and dynamic expert arbitration already address context-dependent combinations. FFORMA learns a feature-to-weight mapping across a collection

of reference series [1]. Arbitrated dynamic ensembles learn and combine expert-competence estimates [2]. These precedents prevent interpreting the existence of contextual weighting as a new general idea. The proposed experiment concerns one particular low-data, within-series memory construction. A small implementation and matching ablations make its mechanism falsifiable; they do not establish priority over earlier ensemble research.

The primary comparison is mean final-block MASE of ContextMemory against RecencyMean over all fifteen selected cases. Configuration values are fixed in the released implementation; this is a retrospective experiment, not a prospectively registered trial or an independently selected confirmation corpus. An unfavorable mean difference provides no aggregate benefit in this experiment. It does not establish that every possible contextual rule is ineffective. Individual case outcomes, interval width and empirical coverage remain visible so that a favorable subset cannot replace the aggregate question.

The paper separates two questions that would otherwise be easy to conflate. The fixed rule asks whether context matching helps when the experts, memory and competence transformation are held constant. The later selection rules ask whether a finite historical comparison can choose a useful allocation before an outer target is exposed. A favorable answer to the second question cannot repair an unfavorable answer to the first. It also cannot identify which selected component caused the change.

The reported units of evidence are equally explicit. The original selected cases are development examples with one final block each. The external collections contain multiple eligible origins, summarized first within a series. An origin is a location at which a forecast is issued, a lead is a position in its future block, and a method-origin path contains all of those leads. These are different denominators. Complete paths permit detailed error inspection, but their individual values are not interchangeable independent replications. The resulting claim is about the stated finite comparisons and their paired series summaries.

The contributions are threefold: (i) an explicit five-expert finite-memory construction with source-resolved context and error definitions; (ii) temporally validated configuration selection and an exhaustively solved regularized simplex fit with declared information boundaries; and (iii) complete two-collection forecast paths and retrospective paired-series uncertainty that retain unfavorable component and comparator results. These are contributions of specification and bounded evaluation, not claims of priority for contextual weighting or stacking.

2 Related work and the scope of this comparison

2.1 Combination, weight estimation and stacking

Bates and Granger [8] used past forecast errors to allocate weights to two competing forecasts. Their example establishes an important distinction for the present study: a combination can benefit from complementary errors without either component being universally best. It does not establish that estimating more flexible weights is beneficial with a small memory. Our shared-expert comparisons therefore ask whether a particular allocation procedure improves prediction, rather than treating the presence of multiple experts as evidence for contextual learning.

Clemen's review [9] documents both the practical interest in combining forecasts and the robustness of simple combinations. This motivates keeping EqualMean visible even when it is not the primary mechanism-isolating control. An estimated procedure should not be judged only against a deliberately weak single expert. Conversely, a gain over equal weights cannot identify why the allocation improved: recency, feature matching, temperature, regularization and historical selection are separate changes in the present implementation.

Timmermann [10] organizes the combination problem around error dependence, instability and weight-estimation uncertainty. This provides a more useful explanation for our design than a generic claim that ensembles reduce variance. All five experts receive the same observations, so their errors can be strongly related. The study does not assume independent experts, nor does the use of five columns create five independent replications. The Gram matrix used by stacking retains cross-products rather than replacing dependence by a diagonal approximation.

Smith and Wallis [11] analyze finite-sample estimation error as an explanation of the forecast-combination puzzle. In our setting, a maximum of twelve remembered origins and five expert coefficients makes that concern concrete. A flexible allocation can fit an unstable relationship among historical errors. Our uniform floor and positive regularization restrict that flexibility; neither device proves that its estimation cost is smaller than its predictive benefit. The observed unfavorable fixed-context comparison is consistent with that concern but does not identify its cause.

Wolpert's stacked generalization [12] learns from predictions made on observations not used to train the corresponding first-stage predictors. This information separation is central to the current comparison. Predictions used to assess a weight rule must have been issued before their targets became available. Our temporal blocks implement that separation for the stated horizon. They do not reproduce arbitrary shuffled folds, and the origin schedule remains part of the estimator rather than an interchangeable evaluation detail.

Breiman's stacked regressions [13] use cross-validation predictions and nonnegative regression coefficients to construct combinations. This is direct prior art for learning constrained linear combinations from held-out errors. Our stacking candidate additionally restricts coefficients to sum to one and penalizes distance from equal weights. Its historical absolute-error selection criterion also differs from its squared-error fitting objective. These differences define an implementation to evaluate; they do not support a priority claim for nonnegative stacking itself.

Hansen's least-squares model averaging [14] selects weights using a Mallows criterion and proves asymptotic properties for its specified model class and assumptions. That theoretical result does not transfer to our short-memory estimator by analogy. We use a different criterion, fitted experts, a capped origin schedule and dependent temporal records. The present theoretical statements are instead finite algebraic properties: uniqueness of the regularized quadratic solution, temporal measurability, convex quantile ordering and a limited absolute-error inequality.

Raftery, Kárný and Ettler [15] develop dynamic model averaging with evolving model parameters and model uncertainty. Their work makes clear that time-varying allocation is established research. Our procedure has no equivalent latent-state probability model and does not claim to approximate that algorithm. We retain a small set of realized expert errors and recompute weights at the requested prefix. Replaying those fits is distinct from a native recursive implementation of dynamic model averaging.

2.2 Features, conditional performance and current alternatives

FFORMA [1] learns a feature-dependent mapping to forecast weights across reference series. Arbitrated forecasting ensembles [2] estimate expert competence and use it to form adaptive predictions. Both are more relevant conceptual comparators than unrelated large models because they address the allocation mechanism. Our recurrence rule estimates relevance from completed origins within one series rather than fitting a global competence model. A fair empirical comparison would need the original training resources, tuning rules, expert roster and information boundaries of each alternative; we have not executed it here.

The broad review by Wang and colleagues [16] places forecast combination within a literature spanning statistical weighting, optimization and learning. The current study occupies a narrow

part of that landscape: small local expert sets, a finite memory and complete historical target blocks. It does not cover hierarchical reconciliation, cross-series representation learning or all combinations of probabilistic distributions. Describing the procedure as better than all published forecast-combination methods would therefore overstate both the comparator roster and the empirical population.

A recent alternative is corrected forecast combinations by Liu and Vasnev [17], whose September 2026 revision considers predictable serial dependence in combined forecast errors. Their correction uses an observable previous error rather than treating all remaining combined error as innovation. This offers a concrete candidate for another experiment, particularly when a completed residual remains relevant at the next query. It has not been implemented or scored in this study. Its results cannot be claimed as our results, and our outcomes do not establish superiority over it.

Giacomini and White [18] distinguish conditional predictive ability from unconditional average performance. This distinction matters because our hypothesis concerns context, while our headline outcome is an average over admitted series. A lower aggregate MASE does not establish that the four context features identify which expert will perform best next. The reported experiment supplies matching controls and selection records, but no conditional predictive-ability test or causal identification of the feature mechanism. Such a test would require an explicit design and suitable forecast records.

2.3 Temporal evaluation and empirical populations

Tashman [19] treats splitting rules, forecast origins, rolling windows and coefficient updates as explicit choices in forecasting evaluation. Our source reconstruction, prefix cap and nonoverlapping outer target blocks follow that need for a fully specified experiment. Three eligible origins provide more than one temporal location for many series, but remain a limited view of a potentially long process. They do not demonstrate behavior under every change in frequency, update policy, horizon or seasonal lag.

Bergmeir and Benítez [20] examine how time-series prediction and overlapping lagged observations complicate evaluation. Their work motivates stating the available information rather than applying a generic cross-validation label. In our construction, an inner training block is admitted only after its last target arrives. This removes direct future-target use at the query. It does not remove dependence between neighboring observations, overlapping fitting prefixes or related series, and it is not presented as proof that all uncertainty calculations become independent.

Bergmeir, Hyndman and Koo [21] show that ordinary cross-validation can be valid for autoregressive prediction under specified error conditions. We therefore do not claim that shuffled cross-validation is universally invalid for time series. Our chronological procedure addresses a different practical contract: every candidate is evaluated with information that would have been observable before its own validation target. That contract remains useful even when a separate theorem would justify another design for another model class.

The Monash archive [22] makes many forecasting collections available in a common format. We identify its weekly and monthly releases by exact source bytes, individual sequence digests and eligibility decisions. This supports checking which series were actually included. It does not turn an archive into a random sample of all operational forecasting tasks. Shared subject matter, related source systems and exclusions can still make an exchangeable-series approximation optimistic.

The M3 study [23] and M4 competition [24] concern their own competition protocols and populations. Our re-use of monthly and weekly sequences changes the available prefix, origin schedule, comparator roster and aggregation rule. Thus, a method with lower MASE in our table has not earned an official competition placement. We report the two collections separately,

disclose exclusions and retain AutoTheta and AutoETS so that the proposed combinations face automatic statistical alternatives as well as matching ablations.

2.4 Scores, uncertainty and claims

Gneiting and Raftery [25] explain why probabilistic evaluation should reward useful concentration while penalizing failures to contain outcomes. Their interval-scoring perspective motivates retaining MSIS beside empirical coverage. A wide Gaussian interval can cover most targets and still provide a poor interval score. In the present comparison, the combination methods' point gains and interval losses are therefore jointly material. Coverage alone is not treated as a quality ranking or a calibration certificate.

Demšar [26] discusses comparisons of learning algorithms over multiple datasets, where the sampling unit differs from individual within-dataset observations. That distinction motivates our series-level summaries, but the paper's classifier-comparison recommendations are not automatically a theorem for this forecasting design. We do not manufacture a larger sample by counting every lead and every remembered fit independently. Our bootstrap preserves all admitted origins of a resampled series and remains conditional on a cross-series exchangeability approximation.

Diebold's discussion of predictive-accuracy tests [27] emphasizes the need to match a comparison to the forecasting question. Our paired-series intervals are not Diebold–Mariano tests on a long stationary loss-difference sequence. They are a descriptive resampling analysis of series summaries after the outcomes were available. Keeping that distinction visible avoids attaching a familiar test name to a different estimand or interpreting an interval that excludes zero as prospective population confirmation.

These relationships define the contribution claimed here. The implementation combines established ingredients in an auditable finite-memory experiment, makes its nested information order explicit, and supplies complete outcomes for specified collections. The favorable results concern temporal selection relative to its recency control. The unfavorable fixed-context, monthly AutoTheta and interval comparisons remain part of the same claim. The literature supplies better questions and stronger candidate comparisons; it does not authorize a broader conclusion than the experiment actually tests.

3 Problem formulation and available information

Let the observed prefix be $y_{1:n}$, the forecast horizon be H and the seasonal lag be m. The final targets $a_{1:H}$ are used only for evaluation. Historical training origins belong to

$$\mathcal{O} \subseteq \{o : o \geq o_{\min}, \quad o + H \leq n\}, \qquad o_{\min} = \max(2m, 2H + 8, 24). \tag{1}$$

At most twelve distinct origins are selected as the unique integer values of a twelve-point equally spaced grid from $o_{\min}$ to $n - H$, with nonnegative grid values truncated to integers. An empty eligible interval supplies no memory. Every expert at origin o receives $y_{1:o}$, and its loss uses $y_{o+1:o+H}$. The requirement $o + H \leq n$ ensures that all remembered targets have arrived before the final query. An origin near the end of the prefix is not eligible merely because its first future target is known; its complete scored horizon must be known.

The selection does not consume the final test block. Generated source definitions and admitted real-data CSVs reconstruct the original equally spaced histories. Every available released display sample is checked at its original time index, the original prefix length is verified, and the reserved targets are checked to their released four-decimal tolerance. A display-decimated replay array

is explicitly rejected as a consecutive fitting sequence. This is necessary because changing observation spacing changes both context statistics and the seasonal error scale.

The forecasting task is univariate. No future promotion flag, calendar regressor or external covariate is supplied to these combination methods. A source observation is one element of the admitted ordered sequence; the protocol does not infer a new sampling frequency from a display layout. The horizon counts future observations in that sequence, and the seasonal lag counts observed positions. These counts are dimensionless indices, while a point forecast, target and interval endpoint have the source series' measurement unit.

Within an origin, every method is evaluated against the same future block. The historical scale has that same measurement unit, making a valid MASE ratio dimensionless. A zero scale cannot be repaired by looking at the future block without changing the task. Missing outcomes, irregular observation spacing and undefined denominators therefore require explicit handling before one can generalize this task to another dataset. The present runner uses admitted finite sequences and records its exclusions; it does not silently interpolate absent observations or relabel an incomplete block as a shorter successful forecast. The common-prefix cap in the external experiment is specified separately from the original selected-case design.

Table 1 collects the mathematical symbols and their units. The generic unit U means the admitted series' measurement unit; the paper does not infer a physical SI unit when source provenance does not specify one. Observation counts and positions describe the supplied sequence's sampling grid.

Table 1: Notation for observed data, context features, causal selection, convex fitting and paired-series evaluation, with measurement units distinguished from dimensionless quantities.

Symbol	Meaning	Unit
y_t	Observation at sequence position t.	U
n, H, m	Prefix length, horizon length and seasonal lag.	observations
a_j, p_{okj}	Reserved target and expert point forecast at a stated lead.	U
$\mathcal{O}_n, o_{\min}$	Outer-prefix memory grid and its first permitted origin.	index set; index
o, v, q	Historical origin, inner validation origin and generic information cutoff.	index
J, K	Number of remembered origins and number of experts; $K = 5$.	counts
R, P, w	Recent window, preceding window and their shared length.	U; observations
d	Half-window length used by the context mean contrast.	observations
$s, \widehat{\sigma}$	Context-window standard deviation and expert residual standard deviation.	U
f_a	Context feature coordinate, including normalized contrasts and cycle correlation.	dimensionless
s_o, s_m	In-prefix training normalization and final seasonal error scale.	U
L_{ok}	Mean absolute expert error at a remembered origin after normalization.	dimensionless
r_j, a_j^{loc}	Recency weight and context-and-recency kernel weight.	dimensionless
D_j, b_D	Context distance and kernel bandwidth.	dimensionless
c_k, g_k, ℓ_k	Competence estimate, recency loss and local-context loss.	dimensionless

Symbol	Meaning	Unit
ρ, τ, ε	Shrinkage, softmax temperature and uniform-floor fraction.	dimensionless
$v_k, w_k, \widetilde{w}_k$	Unfloored allocation, fitted simplex allocation and post-fit floored allocation.	dimensionless
$\mathcal{F}_q$	Information supplied by observations through q.	information set
$\mathcal{T}_v, \mathcal{V}_n$	Completed training-origin set and eligible validation-query set.	index sets
$S_n(c)$	Mean scaled absolute validation error for candidate c.	dimensionless
E_{vhk}, A	Normalized signed expert error and its row-stacked matrix.	dimensionless
N_e	Number of error rows in a stacking fit, equal to JH.	rows
G, λ	Error Gram matrix and positive quadratic regularization coefficient.	dimensionless
u, Δ_K	Uniform allocation and the nonnegative unit-sum simplex.	dimensionless
B, M_B	Active expert set and its regularized Gram matrix.	set; dimensionless
x, z	Solutions of the two linear systems used on an active face.	dimensionless
$F(w)$	Regularized squared-error fitting objective.	dimensionless
ℓ_n, a, b	SES terminal level, SES coefficient and Theta linear slope.	U; 1; U/observation
$\Phi^{-1}(q)$	Standard-normal quantile at probability q.	dimensionless
s, N	External-series identity and number of admitted series in one collection.	index; count
$\bar{L}_s(A)$	Within-series mean loss for method A over its admitted outer origins.	dimensionless
$d_s(A, B)$	Paired series loss difference for methods A and B.	dimensionless
$\bar{d}, \bar{d}^*$	Collection mean difference and its paired-series resampled counterpart.	dimensionless
S_i^*, Q_p	Resampled series identity and empirical quantile at probability p.	index; same unit as argument
α	Interval miscoverage, fixed at 0.2 for the reported MSIS.	dimensionless
$[x]_+$	Positive part, $\max(x, 0)$.	same unit as x

4 Experts and contextual memory

The fixed expert set is SeasonalNaive, SES, Holt, additive HoltWinters and nonseasonal finite-history Theta. Their parameter grids and Gaussian residual quantile rules belong to the implementation. They are shared unchanged by the proposal and both ablations. Seasonal continuation respects an incomplete terminal cycle; Theta includes the finite-history SES correction [3]. No expert is removed after observing its final-block error.

The context uses recent and preceding windows of length $w = \min(\lfloor n/2 \rfloor, \max(8, \min(2m, 48)))$. Let $R = y_{n-w+1:n}$ and $P = y_{n-2w+1:n-w}$, let s be the population standard deviation of their

concatenation, and set $d = \lfloor w/2 \rfloor$. For $w \geq 2$ and $s \geq 10^{-12}$, the first three features are

$$f_1 = (\overline{R_{w-d+1:w}} - \overline{R_{1:d}})/s, \tag{2}$$

$$f_2 = (\overline{R} - \overline{P})/s, \tag{3}$$

$$f_3 = \log\{(\mathrm{sd}(\Delta R) + 10^{-6}s)/(\mathrm{sd}(\Delta P) + 10^{-6}s)\}. \tag{4}$$

The first is a half-window mean contrast, not an OLS slope. The fourth is the Pearson correlation of the last two length-m cycles when $m \geq 2$, $n \geq 2m + 2$ and both cycle standard deviations exceed 10^{-12}; otherwise it is zero. Every feature is clipped to $[-4, 4]$. If the length or scale condition fails, all four features are zero. Population standard deviations use divisor equal to sample length.

For each remembered origin o and expert k, the competence target is a normalized multi-step absolute loss

$$L_{ok} = \frac{H^{-1} \sum_{j=1}^{H} |y_{o+j} - p_{okj}|}{s_o}, \tag{5}$$

Here s_o is the mean absolute lag-$m\prime$ difference in the prefix, with $m\prime = \max(1, \min(m, o-1))$. If that mean is at most 10^{-12}, the training-loss scale is $\max(\mathrm{sd}(y_{1:o}), 10^{-12})$. This training fallback differs from the final evaluation rule, which reports unavailable MASE when the final scale is at most 10^{-12} in source units. No scale is estimated from the final held-out block. The loss array and historical contexts are cached for a fitting history; caching changes runtime rather than the available information.

For remembered origins indexed $j = 1, \ldots, J$, recency weights are $r_j = 2^{-(J-j)/6}$. Distance to the query is $D_j = \sqrt{\frac{1}{4}\sum_{a=1}^{4}(f_{ja} - f_{\mathrm{query},a})^2}$, bandwidth is $b_D = \max(\mathrm{median}_j\, D_j, .25)$, and local weights are $a_j^{\mathrm{loc}} = r_j \exp(-D_j^2/(2b_D^2))$. Expert competence is

$$c_k = .75\frac{\sum_j a_j^{\mathrm{loc}} L_{jk}}{\sum_j a_j^{\mathrm{loc}}} + .25\frac{\sum_j r_j L_{jk}}{\sum_j r_j}. \tag{6}$$

Weights before the uniform floor are normalized values of $\exp(\max\{-(c_k - \min_l c_l), -700\})$. RecencyMean uses only the recency-weighted loss average, without context weighting or shrinkage. EqualMean, and any fit without remembered origins, uses 1/5 directly. These settings are fixed; they are not retuned after inspecting the final blocks.

4.1 Convexity and its limited consequence

Let $v_k \geq 0$ denote the normalized competence allocation before the uniform floor, so that $\sum_k v_k = 1$. The final weight is

$$w_k = (1 - \varepsilon)v_k + \frac{\varepsilon}{K}, \qquad \varepsilon = 0.05, \quad K = 5. \tag{7}$$

Thus $w_k \geq 0.01$ and $\sum_k w_k = 1$. If every input expert has ordered quantiles $q_{k,.1} \leq q_{k,.5} \leq q_{k,.9}$ at a lead, then

$$\sum_k w_k q_{k,.1} \leq \sum_k w_k q_{k,.5} \leq \sum_k w_k q_{k,.9}. \tag{8}$$

This follows by multiplying each input inequality by a nonnegative weight and summing. It proves column order only. An average of marginal quantiles generally differs from the quantile function of a predictive mixture. No calibration or conditional-coverage guarantee follows from this construction.

These are statements about the allocation at a fixed origin; they do not provide an oracle allocation for a future observation or certify generalization.

The context features and normalized losses aim to make expert allocation insensitive to a positive change of units and a common level shift. The numerical implementation includes clipping and degeneracy thresholds, so structural invariance is tested away from those thresholds rather than claimed without qualification for every floating-point input. Independent tests also require finite paths, positive normalized weights and temporally complete historical targets.

4.2 Matching ablations

RecencyMean removes context-distance weighting and retains the experts, completed origins, recency decay and competence-to-weight transformation. It tests whether the contextual component adds value to an otherwise similar recent-loss combination. EqualMean removes estimated competence preferences and assigns a uniform allocation. All three compute their current expert paths from the same original prefix.

Comparing only with SeasonalNaive would not isolate the proposed mechanism: a gain might be due to combining experts rather than context similarity. Comparing only with EqualMean would similarly confound context with loss adaptation. The recency-only ablation is therefore the primary comparator. The equality of expert rosters, information sets and target blocks is part of the experimental contract, not a post hoc interpretation of whichever comparison is favorable.

An additional comparison would be required to isolate the effect of historical selection itself: the selected rule would need to face a corresponding fixed configuration with the same fitting objective and information budget. The current control comparisons remain useful, but cannot attribute every selected-method gain to the context features.

5 Evaluation protocol and corrected selected-case results

The corrected experiment uses fifteen selected cases, eleven generated and four real excerpts, with one final forecast origin per case. For the three combinations, all fitting paths are freshly computed. Other methods in the surrounding atlas include historical frozen forecasts and are not needed to isolate the context-versus-recency question. Their leaderboard positions do not substitute for the matching-expert ablation.

The first calculation incorrectly used display-decimated history arrays as consecutive inputs for nine cases. Its predictions, scales and aggregate values are invalidated. Reconstructing original histories fixes that defect. Every displayed observation matches its original indexed source within 0.000051, and the final targets also match. The invalidated artifact remains identifiable in Git so the correction is auditable; it is not pooled with the corrected experiment or counted as a replication.

The primary metric is MAE divided by the in-prefix mean absolute seasonal difference [4]. This denominator is common to every final-block method within a case. It is not the error of a separately forecast future SeasonalNaive path. Seasonal scales at or below 10^{-12} yield unavailable scaled metrics. Coverage and mean interval width are also retained. Mean scaled interval score supplies an additional descriptive measure without implying an entire predictive distribution has been validated.

5.1 Selected-case results

Table 2: Matching-expert context and recency ablations on the same selected final blocks. All cases, including unfavorable context-weighting outcomes, remain in the reported aggregate.

Case	ContextMemory	RecencyMean	EqualMean
BRKV_level_shift	0.7156	0.7313	0.8425
CHAO_mackey	1.8904	1.8923	1.9291
CTRL_white_noise	1.0934	1.0754	1.1244
EXOG_promo	1.6955	1.7113	1.8232
HETV_garch	3.8878	3.9063	3.8056
INTM_demand	0.0938	0.0867	0.1132
LMEM_fractional	2.1341	2.1377	2.2060
MSEA_daily_weekly	0.6951	0.6857	1.2424
REAL_electricity	0.6944	0.6937	0.7674
REAL_m4_daily	0.2912	0.2939	0.2928
REAL_m4_hourly	1.6003	1.6067	3.2852
REAL_pm25	0.4174	0.4174	0.4173
RWLK_noise	4.9847	4.9333	5.0806
SEAS_hourly	1.0072	1.0070	4.4859
TRND_seasonal	0.5557	0.5545	1.0389
Mean	1.4504	1.4489	1.8970

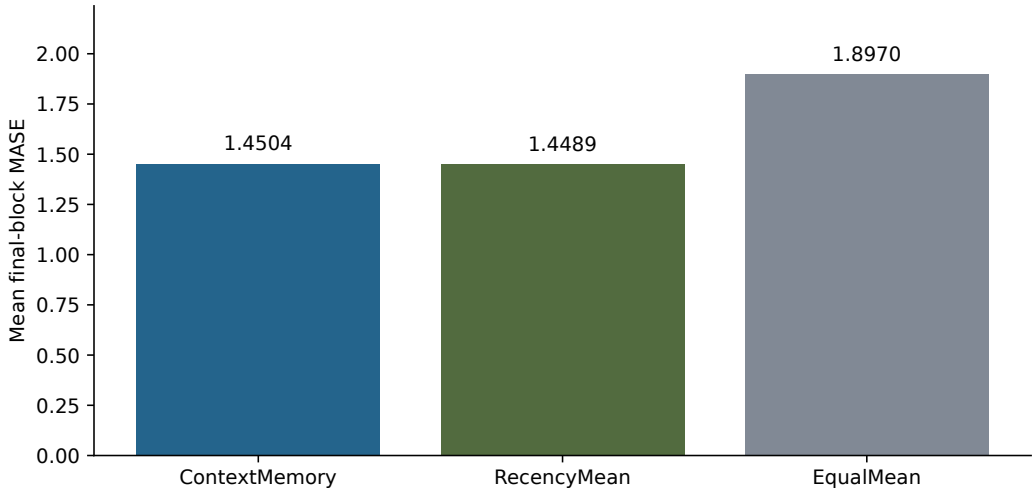


Figure 1: Mean MASE of the fixed combinations on all selected final blocks. ContextMemory is slightly worse than RecencyMean at the declared primary criterion. Bar heights are point summaries; no sampling uncertainty is shown because one selected origin per case does not supply a justified population interval.

Figure 1 summarizes the matching ablation in Table 2. ContextMemory has mean MASE 1.4504, compared with 1.4489 for RecencyMean and 1.8970 for EqualMean. The contextual component fails its aggregate improvement criterion. The estimated recent-loss combinations perform better than the equal average on this collection, but that difference does not establish a benefit from the additional context kernel.

Figure 2 retains the individual directions. The aggregate difference is small relative to variation across selected mechanisms. It is still unfavorable at the criterion that was actually declared. Rounding both means to two decimal places would conceal that outcome. No significance claim,

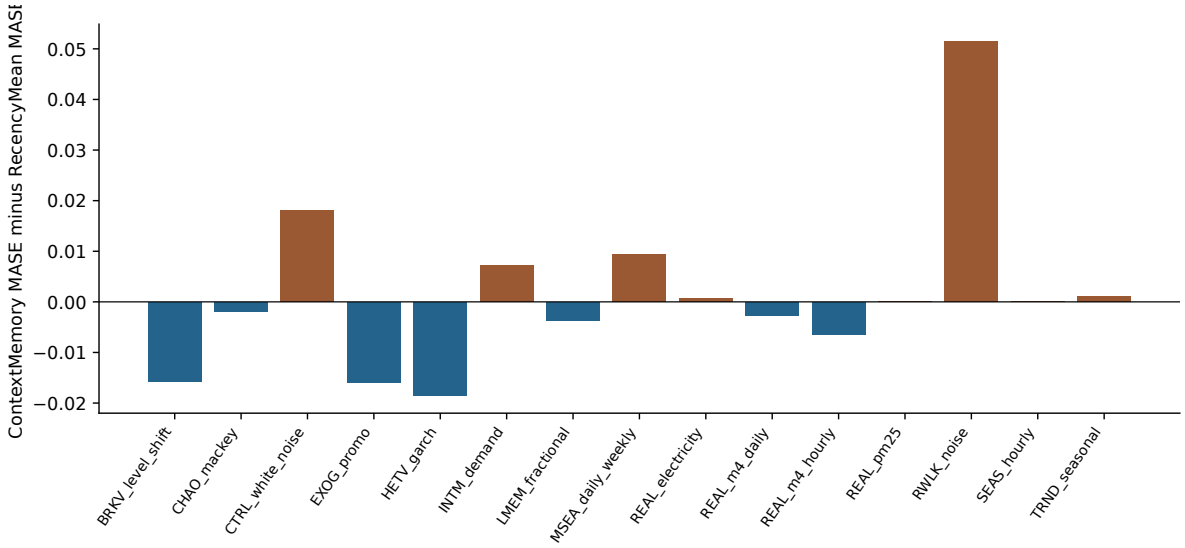


Figure 2: Per-case MASE difference between contextual and recency-only weighting. Each bar is one final block. Both signs are retained; selecting only negative bars would replace the declared aggregate question with a post hoc favorable subset.

equivalence conclusion or general impossibility result is attached to the difference. Full-precision prediction paths and expert weights remain available so the cases that help and hurt can be inspected without reconstructing an aggregate from a selected table.

6 Proposed method and theoretical derivation

The fixed context-versus-recency comparison tests one particular parameter setting. Two additional methods allow a historical validation calculation to choose a setting before an outer forecast. They retain the same five experts, four context features, completed-origin memory and quantile-column combination. Their improvements must not be attributed exclusively to the context kernel, because temperature, shrinkage and selection also change.

For every remembered validation origin v, only training origins o satisfying $o + H \leq v$ are eligible. At least three such origins are required. The candidate is fit on those completed earlier blocks, predicts the block beginning at v, and receives its mean absolute error divided by a scale calculated from the prefix at v. The final candidate score averages these eligible validation-block losses. The outer target is never an input to candidate fitting or selection. If no validation origin qualifies, the method uses the original RecencyMean rule.

ContextMemoryCV considers context shrinkages $\{0, .5, 1\}$ and temperatures $\{.15, .5, 1\}$, plus the original RecencyMean and EqualMean controls. Shrinkage one removes the context kernel, so selecting this method does not imply that contextual similarity helped. All settings retain recency half-life six, the original bandwidth rule and the 5% uniform allocation floor. The first configured minimum wins an exact score tie. After selection, the chosen rule is fit on all completed memory blocks in the outer prefix.

ValidationMix adds four regularized convex stacking candidates with penalties $\{.01, .1, 1, 10\}$. They fit normalized signed expert errors and retain the same five-expert roster. The detailed objective, active-face solution, uniform floor and distinction between fitting and selection losses are derived below. These candidates extend the eleven-rule ContextMemoryCV roster to fifteen rules; established convex stacking is not claimed as a new general idea.

This section gives a complete mathematical interpretation of the implemented selection procedure. The proposal is the finite configuration tested in this paper, not a new general principle of stacking. Its input is one contiguous observed prefix, a positive seasonal lag and a positive forecast horizon. Its output includes three quantile columns, an expert allocation, the selected configuration, historical origins and validation records. There is no argument through which the caller can supply the outer reserved targets. Rejecting invalid or nonfinite input is part of the computation, rather than a reason to omit a difficult case after seeing its score.

6.1 Three distinct information boundaries

Write $\mathcal{F}_q = \sigma(y_1, \ldots, y_q)$ for the information supplied by the series through query q. The memory grid $\mathcal{O}_n$ is fixed from the length of the outer prefix and the declared lag and horizon. A validation query $v \in \mathcal{O}_n$ may use only the completed subset

$$\mathcal{T}_v = \{o \in \mathcal{O}_n : o + H \leq v\}. \tag{9}$$

The rule requires at least three members before scoring that query. The validation origin's own context is computed from its prefix; its expert forecasts are computed from that same prefix. Historical loss rows and normalized signed errors used to fit its candidate weights come only from $\mathcal{T}_v$. The candidate's validation outcome is then formed from the block after v. That block is a target at this inner stage, although it is already observed at the final outer origin n.

Consequently, the whole inner validation calculation is available when the outer fit is issued. Provided the expert implementations are deterministic functions of their supplied prefixes and fixed configuration, the final weights are $\mathcal{F}_n$-measurable. This is an information-order statement. It does not assert that historical targets are statistically independent of their prefixes or of each other. Nor does it assert that a chronological split makes a changing process stationary. It simply specifies which observations can participate in each role and rules out use of the outer reserved block in fitting or selection.

There is a subtle limitation to the grid that must remain explicit. The origin grid is constructed from the supplied outer-prefix length. We then filter that grid at each historical query; we do not claim to recreate a historical service that would have selected its own independent grid knowing only its then-current length. The selected errors are chronologically available, but their sampling schedule is an offline evaluation design. A deployment requiring exact historical service reconstruction should issue and retain each prediction prospectively instead of interpreting this capped grid as such a log.

6.2 Candidate scores and deterministic selection

Let $\mathcal{V}_n$ be the validation queries with at least three completed training blocks. For a candidate c, let $w^{(c,v)}$ be its allocation fitted using those blocks, and let E_{vhk} be expert k's signed error at query v, lead h, divided by the query-prefix training scale. The recorded criterion is

$$S_n(c) = \frac{1}{|\mathcal{V}_n|} \sum_{v \in \mathcal{V}_n} \frac{1}{H} \sum_{h=1}^{H} \left| \sum_{k=1}^{K} E_{vhk} w_k^{(c,v)} \right|. \tag{10}$$

Every candidate is scored on the same eligible queries. The selected configuration is the first configured minimizer of this criterion. Selection is then followed by a new weight fit using every completed memory block in the outer prefix. Historical validation estimates the performance of the allocation rule; its particular inner coefficient vector is not copied as if it were necessarily the correct final vector.

If no validation query qualifies, there is no historical selection estimate to minimize. The implementation records unavailable candidate means and explicitly selects the original recency-only candidate. That fallback can still allocate weights from any completed outer memory that exists. If the memory itself is empty, the allocation is uniform. Distinguishing these two cases matters: absence of an eligible validation split is not identical to absence of historical expert losses. Neither case justifies inventing a favorable validation score or replacing the fallback with the best outer-test method.

For a contextual candidate, let g_k and ℓ_k be recency-weighted and context-and-recency-weighted normalized losses. Its shrinkage ρ and temperature τ determine

$$c_k = (1-\rho)\ell_k + \rho g_k, \qquad v_k = \frac{\exp\{\mathrm{clip}[-(c_k - \min_j c_j)/\tau, -700, 0]\}}{\sum_l \exp\{\mathrm{clip}[-(c_l - \min_j c_j)/\tau, -700, 0]\}}. \tag{11}$$

Subtracting the minimum does not change exact softmax ratios and ensures that at least one unnormalized term is one. Clipping bounds numerical underflow in the remaining terms. Temperature changes concentration; shrinkage changes which loss estimate is concentrated. At $\rho = 1$, the context distance has no effect. Thus selecting a configuration whose name belongs to the contextual family is not evidence that its kernel contributed to prediction quality.

6.3 Regularized stacking on a simplex

For stacking, arrange all completed training errors into a matrix A with $N_e = JH$ rows and $K = 5$ columns. Let $u = \mathbf{1}/K$ and $G = A^\top A/N_e$. The unfloored objective can be written

$$F(w) = w^\top G w + \lambda(w-u)^\top(w-u), \qquad w \in \Delta_K = \{w \geq 0 : \mathbf{1}^\top w = 1\}. \tag{12}$$

The Gram term preserves correlations and relative signs of expert errors. A candidate can therefore benefit from cancellation even when no single expert has uniformly small absolute loss. The penalty trades that fitted cancellation against proximity to equal allocation. Its fixed positive grid is part of the evaluated algorithm. Changing the penalty grid, expert set, memory schedule or error normalization produces another configuration, even if the public method label were left unchanged.

Because G is positive semidefinite and $\lambda > 0$, the Hessian is

$$\nabla^2 F = 2(G + \lambda I) \succeq 2\lambda I. \tag{13}$$

The objective is strictly convex, and the nonempty compact simplex supplies a unique exact minimizer. This follows from standard convex optimization [28]; it is not a new statistical generalization theorem. Strict convexity does not imply good future forecasts, well-estimated covariances or correct interval coverage. It gives a well-defined fitting problem whose numerical answer can be checked independently of those empirical questions.

Five columns make exhaustive face enumeration practical. For a nonempty active set B, define $M_B = G_{BB} + \lambda I$, $x = M_B^{-1}\lambda u_B$ and $z = M_B^{-1}\mathbf{1}$. The equality-constrained stationary point on that face is

$$w_B = x + \frac{1 - \mathbf{1}^\top x}{\mathbf{1}^\top z} z, \qquad w_{B^c} = 0. \tag{14}$$

Positive definiteness makes the denominator positive in exact arithmetic. The implementation solves the two right-hand sides together, rather than explicitly forming an inverse. A face with a coefficient below -10^{-10} or a nonfinite answer is rejected. Accepted coefficients are clipped only within the stated numerical feasibility tolerance and normalized before evaluating the original objective. All 31 nonempty faces are considered, including single-expert vertices, and a feasible equal allocation initializes the objective comparison.

Any simplex minimizer has a nonempty support. Its restriction to the relative interior of that support satisfies the corresponding equality-constrained first-order condition, while a boundary optimum appears on a smaller enumerated face. Therefore evaluating all feasible face solutions recovers the exact constrained optimum under exact arithmetic. In floating-point execution, tolerances and linear-system failures remain numerical concerns. Retaining feasibility, objective and independent reference checks is more informative than treating the absence of an iterative optimizer as an automatic certificate of numerical correctness.

The common uniform floor is applied after fitting:

$$\widetilde{w} = .95w + .05u. \tag{15}$$

The final allocation lies inside the simplex and each expert receives at least .01. It generally does not minimize F; the objective was solved for w, not for its floored transform. Likewise, the candidate is selected by scaled absolute validation error, not by F. These are two separate layers: a squared-error regularized fitting rule within a candidate and an absolute-error historical selection rule across candidates. Reporting both prevents the word optimal from obscuring which optimization was actually performed.

6.4 What convex allocation guarantees

At a fixed observation, convexity of absolute value yields

$$\left| y - \sum_k w_k p_k \right| \leq \sum_k w_k |y - p_k|. \tag{16}$$

The right side is a weighted average of component losses. It is not their minimum, and the inequality does not establish that the combination beats the best expert. The error may still be larger than that of an individual method or an equal average. Averaging over a horizon preserves the inequality for fixed weights, but does not turn historical weight estimation into an oracle for future errors. This limited consequence is useful for interpreting the algorithm without promising an empirical ranking.

For the same reason, positive expert weights preserve the order of already ordered quantile columns, but do not make their weighted columns a calibrated predictive distribution. Our experts' Gaussian residual approximations can be too wide, asymmetric relative to actual errors, or poorly adapted to horizon-specific uncertainty. A weight rule selected for absolute point error has no direct objective encouraging nominal marginal coverage. The observed high coverage and poor MSIS of several combinations are therefore substantive evidence about this construction, rather than a contradiction of convexity.

Finally, normalized features and errors explain why the allocation aims to be stable under a positive affine unit conversion. Away from degeneracy thresholds, translating the series cancels from mean contrasts and errors, while multiplication by a positive scale cancels from the corresponding standard deviations and seasonal error denominators. The expert equations must also transform consistently. The implemented absolute tolerances and finite precision require explicit exceptions near zero scales; we do not infer universal unit invariance from symbolic cancellation alone. Final evaluation retains unavailable MASE for a seasonal scale at most 10^{-12} in source units instead of silently using the training fallback.

The mathematical properties above support deterministic, inspectable fitting. They do not supply the missing independent comparator experiments, prospective confirmation on a deployment population, or an isolated attribution of gains to context. Those remain empirical questions. Complete validation records and retained adverse comparisons make the next experiment more precise: one can change a single allocation ingredient, preserve its information contract, and

evaluate whether the previously observed point gains survive without reusing the new target block for selection.

6.5 Computational and memory requirements

Let K be the expert count, J the number of remembered origins, $V \leq J$ the eligible inner-query count, $d = 4$ the context dimension, and $Q = 3$ the retained quantile count. Write $F_k(n_o, H)$ for the actual fitting and forecasting work of expert k at prefix length n_o. Building the memory requires $\sum_{o \in \mathcal{O}_n} \sum_{k=1}^{K} F_k(n_o, H)$, in addition to context extraction and array construction. For the finite smoothing grids used here, the recursive fits are linear in prefix length with a fixed grid-size multiplier. They are fresh finite-prefix fits, not constant-time native state updates. The final expert paths require another fit on the complete supplied prefix.

Memory stores context rows, expert losses, normalized error rows and all historical quantile paths. Its numerical-array requirement is $O(Jd + JK + JHK + JKHQ)$, dominated by the historical paths when H grows. The local LRU cache retains at most sixteen memory entries; their history-tuple keys also retain observed values. Cache reuse avoids repeating identical calculations but does not turn those values into newly observed information. Each worker has its own process-local cache, so increasing worker count can increase peak memory.

For one contextual candidate and inner query, distance and competence allocation cost $O(Jd+JK)$ after memory construction. Stacking first forms a K-by-K Gram matrix from JH error rows, costing $O(JHK^2)$. Enumerating every nonempty active expert set then has the upper bound $O(2^K K^3)$ for dense face solves. With C_c non-stacking and C_s stacking candidates, an upper bound for the inner allocation work is

$$O\Big[C_c V(Jd + JK) + C_s V(JHK^2 + 2^K K^3)\Big], \tag{17}$$

plus scoring the H-lead paths and the final selected allocation. In this experiment $K = 5$, $J \leq 12$, $C_c = 11$ and $C_s = 4$; the exhaustive solver considers 31 faces. These fixed small values explain why exact face enumeration is feasible here. The exponential dependence on K precludes inferring scalability to a large expert library from this experiment. We report no recovered hardware-normalized latency comparison: retained per-series seconds do not identify the historical machine or the elapsed makespan of parallel execution.

6.6 Final-block scoring and missing scales

Let a_j be the reserved target, p_j the central point forecast, and l_j, u_j the outer quantiles for lead $j = 1, \ldots, H$. Local Gaussian experts use their central forecast and a combination weights those centers. AutoETS and AutoTheta supply point predictions through the reference API mean. For the supplied prefix of length n, set $m' = \max(1, \min(m, n-1))$ and

$$s_n = \frac{1}{n - m'} \sum_{t=m'+1}^{n} |y_t - y_{t-m'}|, \quad \mathrm{MAE} = H^{-1} \sum_j |a_j - p_j|, \quad \mathrm{MASE} = \mathrm{MAE}/s_n. \tag{18}$$

The reference scale uses only the observed prefix, not the held-out block. It is recomputed for each origin, shared across methods at that origin, and then each method's origin scores are averaged within its series. Thus series with more eligible origins do not automatically receive greater weight in a collection summary. MASE compares future error to a historical seasonal-difference scale; it does not by itself establish superiority over a future seasonal-naive forecast [4].

For miscoverage $\alpha = .2$, the central interval score and its scaled form are [25]

$$\mathrm{IS}_{\alpha,j} = (u_j - l_j) + \frac{2}{\alpha}[l_j - a_j]_+ + \frac{2}{\alpha}[a_j - u_j]_+, \tag{19}$$

$$\mathrm{MSIS} = \frac{1}{H s_n} \sum_j \mathrm{IS}_{\alpha,j}, \qquad \mathrm{Cov} = H^{-1} \sum_j \mathbf{1}\{l_j \le a_j \le u_j\}. \tag{20}$$

Here $[x]_+ = \max(x, 0)$. Interval score retains the source unit, while MSIS is dimensionless and penalizes both width and missed observations. Greater empirical coverage is not automatically better: a wide interval can improve coverage while worsening its score. Endpoints include equality in the coverage event.

The scorer rejects nonfinite, crossed or mismatched forecast vectors. With valid vectors it reports unavailable scaled scores when $s_n \le 10^{-12}$. The external runner excludes a series if any required origin lacks a finite scale above that threshold. The absolute threshold has source units and breaks exact affine scale invariance when active; it is not a universal dimensionless criterion for a constant series. This rule differs from the training-loss fallback to prefix standard deviation. The admitted experiment retains this distinction rather than replacing the denominator after observing a result.

7 External evaluation and results on weekly and monthly collections

The sources are the complete Monash M4 Weekly inventory of 359 identities [5] and M3 Monthly inventory of 1,428 identities [6], both redistributed under CC BY 4.0. ZIP, extracted TSF and exact sequence SHA-256 values identify their bytes. Forecast horizon and seasonal lag are $(13, 52)$ for weekly and $(18, 12)$ for monthly. Each series supplies up to three nonoverlapping final target blocks, provided that the prefix has at least $\max(2m, 2H + 8, 24)$ observations. Every method receives the same final 512 contiguous prefix observations, or all observations if fewer are available. The scale uses this supplied prefix, and never the future block.

Sixty-five weekly identities have no eligible origin under that fixed criterion. All 294 eligible weekly series contribute three origins, giving 882 origins. Six exact duplicate monthly sequences retain one representative each; 1,422 monthly series contribute 3,547 eligible origins. Every exclusion and representative remains identifiable. Scores average origins within a series and then weight admitted series equally within each collection. The collections are reported separately; their different units, seasonal scales and origin counts are not silently pooled. This schedule and scaling differ from the original forecasting competitions, so the results do not establish an official competition rank.

Twelve methods are evaluated at every admitted origin: five individual experts, AutoETS and AutoTheta from StatsForecast [7], three fixed combinations, ContextMemoryCV and ValidationMix. Automatic statistical methods use the same available prefix and configured seasonal lag. Their library point forecast supplies the central column and their requested analytic 80% interval supplies the endpoints. This roster contains no freshly evaluated neural, pretrained or globally trained ensemble on the external collections. A method failure stops the calculation rather than deleting a difficult comparator.

Table 3 and Figure 3 summarize both external collections.

ValidationMix has mean MASE .2855 versus .3109 for RecencyMean on weekly series, and .8993 versus .9085 on monthly series. ContextMemoryCV gives .2848 and .8996. AutoTheta gives .3076 and .8472: the temporally selected combinations have lower point error on the admitted weekly collection, while AutoTheta has lower point error on monthly series. These are finite-collection

Table 3: Complete external comparison. W denotes 294 eligible M4 Weekly series; M denotes 1,422 unique M3 Monthly series. MASE and MSIS average origins within series and then series equally; Cov is empirical interval coverage. These are descriptive summaries, not confidence intervals.

Method	W MASE	W MSIS	W Cov	M MASE	M MSIS	M Cov
SeasonalNaive	0.8446	8.1670	0.9326	1.0864	8.8123	0.9342
SES	0.3533	2.3404	0.8926	1.0456	7.1945	0.8534
Holt	0.3806	2.4611	0.8844	1.8040	10.8472	0.8005
HoltWinters	0.5396	3.0778	0.8413	0.9375	6.7112	0.9160
Theta	0.3469	2.3237	0.9041	1.0021	7.0453	0.8700
AutoETS	0.3413	1.8490	0.8713	0.8514	4.1518	0.7324
AutoTheta	0.3076	1.7125	0.8511	0.8472	4.2994	0.7279
ContextMemory	0.3079	2.9468	0.9751	0.9086	6.8846	0.9348
RecencyMean	0.3109	2.9315	0.9742	0.9085	6.8846	0.9348
EqualMean	0.3548	3.3233	0.9702	0.9962	7.2131	0.9333
ContextMemoryCV	0.2848	2.5538	0.9511	0.8996	6.7999	0.9329
ValidationMix	0.2855	2.5477	0.9503	0.8993	6.8208	0.9326

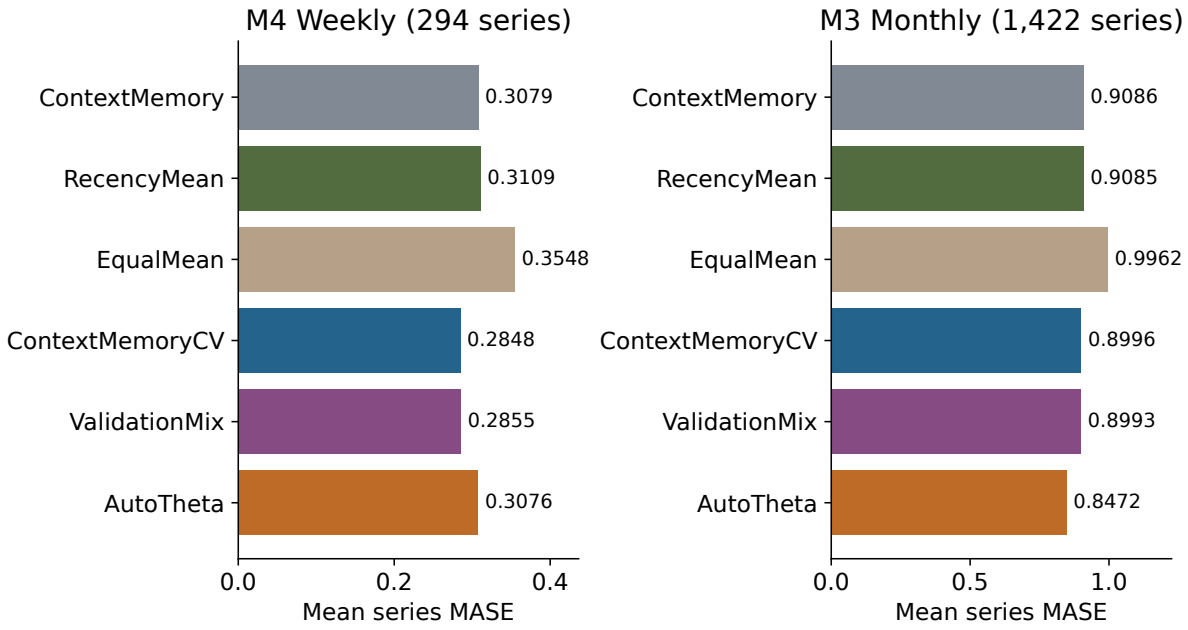


Figure 3: Fixed and temporally selected combinations alongside AutoTheta on each external collection. Bars are mean series MASE without sampling-error bars. The complete twelve-method table retains all individual-expert and automatic-statistical outcomes.

comparisons without an inferential or population-superiority claim.

Uncertainty quality does not follow the point-error ordering. ValidationMix has weekly MSIS 2.5477 and coverage .9503, versus AutoTheta's 1.7125 and .8511. On monthly series the corresponding values are 6.8208 and .9326, versus AutoTheta's 4.2994 and .7279. The combinations' wide Gaussian approximations remain a material limitation at nominal coverage .8. Selecting weights for point error does not calibrate their interval endpoints.

The original fifteen final blocks remain development examples. On those blocks ContextMemoryCV and ValidationMix have mean MASE 1.3974 and 1.4031; the original fixed contextual and recency-only results remain 1.4504 and 1.4489. Their MSIS values are 9.5025 and 9.3428 versus RecencyMean's 9.1076. The changed algorithms therefore improve this descriptive point score while worsening the interval score. Reusing these already examined cases is not independent external confirmation.

8 Statistical unit and exploratory paired uncertainty

An external benchmark collection contains many forecast origins but not an equal number of independent replications. Several origins of one series share most fitting observations, and the forecast leads belong to one jointly generated future block. Treating every lead or origin as an independent replication would make the reported precision depend on how densely the same series was evaluated. We therefore first average the available origins within a series and then give each admitted series equal weight. This choice defines the population summary being described; it is not merely a computational convenience for obtaining a narrower interval.

Let $L_{so}(A)$ be a loss for method A at origin o of series s, and let $\mathcal{O}_s$ contain that series' admitted origins. Define

$$\bar{L}_s(A) = |\mathcal{O}_s|^{-1} \sum_{o \in \mathcal{O}_s} L_{so}(A), \qquad d_s(A, B) = \bar{L}_s(A) - \bar{L}_s(B). \tag{21}$$

The collection mean difference is

$$\bar{d}(A, B) = N^{-1} \sum_{s=1}^{N} d_s(A, B). \tag{22}$$

For MASE and MSIS, a negative value favors A. A signed coverage difference has no universal direction of improvement: increasing coverage by making intervals wider can worsen their usefulness. Coverage differences are retained in the complete evidence, but are not labeled loss reductions.

8.1 Collection admission and the denominator of a comparison

Table 4 identifies the source inventory and the actually admitted comparison. The weekly exclusions are recorded under the combined reason of nonfinite observations or no eligible origin; that category does not support attributing every exclusion to a particular history-length threshold. The monthly exclusions are exact duplicates. All excluded identities remain visible in the numerical index, so the denominator is not reconstructed from successful forecasts alone. A comparison conditioned on eligibility does not describe source series for which the specified fitting protocol could not produce an origin.

The external comparison contains 4,429 origins and twelve methods, yielding 53,148 method-origin forecast paths: 10,584 weekly paths with thirteen leads and 42,564 monthly paths with eighteen leads, or 903,744 individual point predictions. The same future targets are scored for every

Table 4: Source inventories and admitted external evaluation units. Each origin has twelve methods; the weekly and monthly horizons differ. Point values count one scalar per method and lead, not all three quantiles.

Collection	H	Source	Admitted	Excluded	Origins	Point values
M4 Weekly	13	359	294	65	882	137,592
M3 Monthly	18	1428	1422	6	3547	766,152
Total	–	1787	1716	71	4429	903,744

method at an origin. Counting those repeated method-target pairs as independent experimental outcomes would confuse computational volume with inferential sample size. The series-level mean in Equation 22 uses 294 weekly or 1,422 monthly observations, not the number of scalar predictions. The separate fifteen selected development cases are retained for comparison with the original proposal; they are not added to either external collection's inference denominator.

8.2 What was resampled, and what was not

One bootstrap draw samples N complete series identities with replacement and averages their paired differences. The two methods are never sampled independently. All origins already belonging to a sampled series remain together through its within-series mean. M4 Weekly and M3 Monthly are analyzed separately, because pooling their differently selected series and scale denominators would change the estimand. With resampled identities $S_1^*, \ldots, S_N^*$, the bootstrap statistic is

$$\bar{d}^* = N^{-1} \sum_{i=1}^{N} d_{S_i^*}, \qquad C_{.95} = [Q_{.025}(\bar{d}^*), Q_{.975}(\bar{d}^*)]. \tag{23}$$

The computation uses 10,000 local PCG64 draws, linear empirical quantiles and seed 20261105 plus a stable collection-specific offset. Identical resampled indices are used across comparisons within a collection. Batch size bounds memory consumption without changing the random sequence.

This uncertainty analysis was specified after the original outcomes were available. It does not refit a forecaster, alter an expert weight, change a validation schedule, remove an unfavorable series or obtain new external data. Its intervals are unadjusted and exploratory. Under independent representative series and regularity conditions, a series bootstrap can describe sampling variation in a mean difference. The intervals here are not finite-sample distribution-free coverage guarantees. Related industries, shared source processes or other dependence between nominally distinct series can invalidate independent-series resampling. Exact whole-sequence deduplication removes a clear duplication defect, but does not prove that all remaining series are statistically independent.

8.3 Comparisons and their different interpretations

Table 5 reports ValidationMix minus two informative controls. RecencyMean isolates whether a temporally validated combination improves beyond the matching recency-only memory rule. AutoTheta is a separately implemented statistical forecast and supplies a stronger external alternative on several metrics. The controls answer different questions; improvement over the recency ablation cannot be presented as superiority over the complete roster. The complete numerical file additionally retains ContextMemoryCV and AutoETS, all signed coverage comparisons, means, medians and loss-direction counts.

The MASE differences favor ValidationMix over RecencyMean in both collections, and their paired intervals exclude zero under the stated resampling procedure. This is evidence about these

Table 5: ValidationMix minus each control after averaging available origins within each series. Unadjusted exploratory 95% paired-series intervals describe the admitted collection, not population-wide superiority.

Collection	Control	Loss	Difference	95% interval	Lower loss
Weekly	RecencyMean	MASE	−0.025337	[−0.033914, −0.017227]	190/294
Monthly	RecencyMean	MASE	−0.009154	[−0.013176, −0.005173]	589/1422
Weekly	AutoTheta	MASE	−0.022065	[−0.037521, −0.008272]	155/294
Monthly	AutoTheta	MASE	0.052154	[0.032437, 0.073112]	689/1422
Weekly	RecencyMean	MSIS	−0.383808	[−0.431160, −0.334374]	252/294
Monthly	RecencyMean	MSIS	−0.063788	[−0.088692, −0.040255]	589/1422
Weekly	AutoTheta	MSIS	0.835230	[0.650853, 1.038115]	49/294
Monthly	AutoTheta	MSIS	2.521379	[2.364504, 2.673453]	267/1422

measured aggregate differences. It does not identify which component of the selector caused the difference, prove out-of-collection generalization or establish that most individual series improve. On M3 Monthly, only 589 of 1,422 series have strictly lower MASE than RecencyMean, even though the collection mean is lower. Unequal magnitudes of improvements and regressions, together with equal outcomes, can produce that combination. Mean difference and win fraction are different summaries.

Against AutoTheta, the point-error direction depends on the collection. ValidationMix has the favorable weekly mean difference, while AutoTheta has the favorable monthly difference. AutoTheta also has lower MSIS in both collections. Thus the experiment does not identify a universally preferable forecast distribution. It identifies a bounded improvement of a temporally validated combination relative to its recency control, while retaining a stronger alternative for monthly point error and both interval scores.

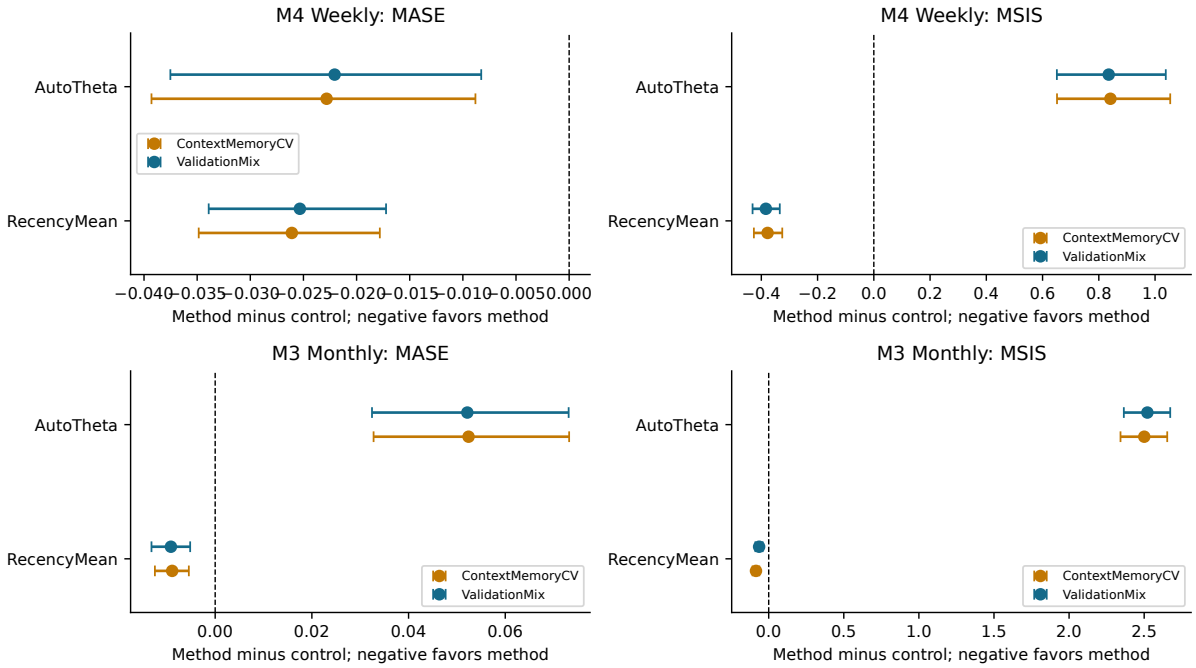


Figure 4: Exploratory paired-series intervals for the two temporally validated combinations. Negative loss differences favor the combination; separate panels retain differences between collections, controls and point versus interval losses.

Figure 4 places both combinations against matching controls, rather than displaying only a

favorable subset. The MSIS panels show that a point-error improvement does not guarantee a better distributional forecast. Figure 5 complements the mean differences with complete series-level direction counts. Equal-loss series remain in the denominator. Removing them would change the displayed fraction and could create an impression of broad individual gains that the complete collection does not support.

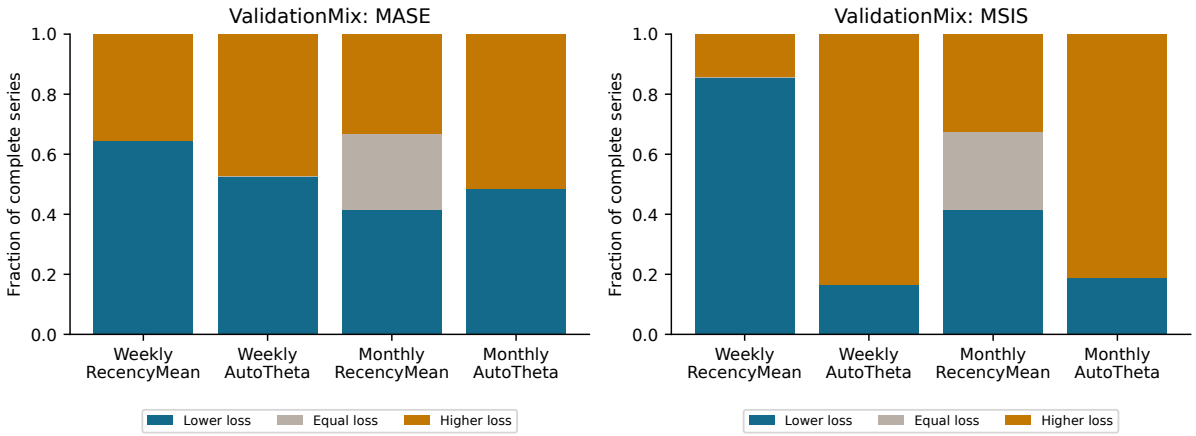


Figure 5: Fractions of admitted series with lower, equal or higher ValidationMix loss after averaging origins. Aggregate mean gains need not imply that most series improve, especially when loss magnitudes are unequal.

8.4 Causality, selection and attribution

The outer forecast at query origin q may select a rule using a historical validation origin o only when its entire target block is already observed:

$$o + H \leq q. \tag{24}$$

Every historical expert fit uses observations available at its own origin. The query's reserved targets enter neither an expert fit nor the selection criterion. Figure 6 depicts this information order; its positions illustrate the constraint and are not a measured timing result. Using a recently started but incomplete target block would leak future information into the selector even if its features were computed from a valid prefix. The completion condition therefore applies to outcomes as well as features.

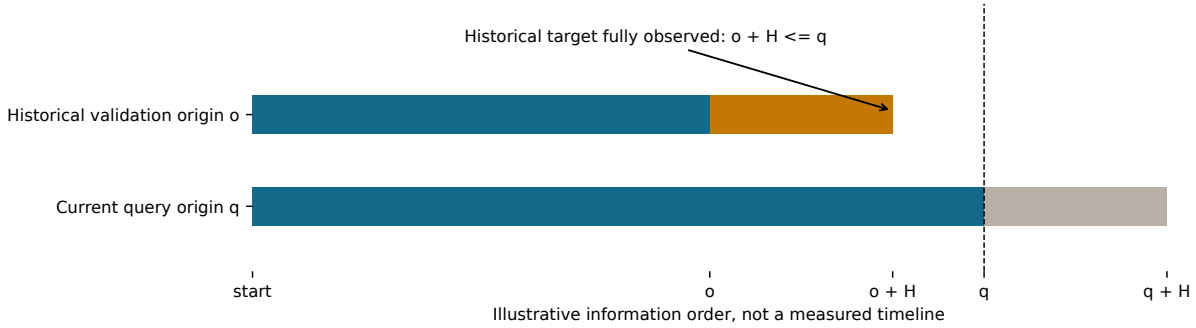


Figure 6: Illustrative information order for historical validation and a current query. A historical target must be completely observed before it can inform selection; the gray current target remains reserved for evaluation.

Algorithm 1 Temporally completed candidate selection. Every expert forecast and validation target used to choose a combination is available before the current reserved target block begins.

Require: Observed prefix $y_{1:n}$, horizon H, fixed candidates $\mathcal{C}$

Construct the fixed historical origin schedule and cache expert paths
for each historical query origin q **do**
    Retain earlier origins o satisfying $o + H \leq q$
    Use the query only when at least three completed origins are available
    for each candidate $c \in \mathcal{C}$ **do**
        Fit its weights from the completed historical memories only
        Score its combined path on the query block using the query's prefix scale
    end for
end for
Select minimum mean validation loss; break ties by the fixed grid order
With no eligible query, use the explicitly defined recency-only fallback
Fit the selected rule's current weights from all completed prefix memories
Return the weighted current expert path without observing $y_{n+1:n+H}$

Algorithm 1 separates the selection protocol from its candidate weight formula. A grid containing contextual softmax weights and regularized stacking does not let either candidate train on its own validation outcomes. The fallback is part of the declared algorithm, not an exclusion of short or inconvenient prefixes after their errors are observed. The final outer block remains reserved regardless of which candidate is selected.

Causal data access prevents one important validity defect, but does not remove selection uncertainty. A fixed finite candidate grid can still overfit a small set of completed historical blocks. ContextMemoryCV and ValidationMix can choose different candidates at different outer origins, and an aggregate gain is the outcome of the complete procedure. It is not an isolated estimate of the context kernel's causal contribution. The recency-only and equal-weight rules provide informative ablations; neither equals a fully crossed experiment in which every stacking, regularization and context choice is independently randomized or disabled while the selection budget is otherwise identical.

The resampling analysis consequently leaves the original scope intact. It adds uncertainty and distributional summaries to frozen forecasts, rather than converting a retrospective comparison into a preregistered trial. Independent code recomputes the reported differences from full-precision audit paths and confirms that two analysis processes produce identical JSON bytes. This supports computational reproducibility. The externally admitted series, fixed temporal protocol and explicit unfavorable controls bound the scientific interpretation of the resulting means and intervals.

9 Discussion and limitations

Context matching has a bias–variance tradeoff. A small within-series memory can contain few genuinely comparable contexts. Kernel weighting can overemphasize those few origins, while shrinkage and a uniform floor can dilute a useful specialist. A context can recur in its four descriptive statistics without recurring in the forecasting relationships that determine competence. The experiment does not identify which explanation caused the observed aggregate failure; that requires new controlled comparisons.

The original selected corpus and one final origin per series provide a bounded diagnostic experiment rather than representative population sampling. The additional weekly and monthly collections

broaden empirical coverage, but eligibility exclusions, source relationships and limited rolling origins still constrain generalization. Training origins share observations and must not be counted as independent final-test replications. Methods also share experts and target series. A dependence-aware uncertainty analysis would need an explicitly designed repeated-origin or cross-series study. Neither the selected-case experiment nor the external collections support a superiority or equivalence statement for a deployment population.

Faithful external FFORMA and arbitrated-ensemble comparators remain necessary before asserting a research contribution beyond this tested construction. A fair study would also report fitting time, cache memory, inference latency and the information available to those comparators. A successful future experiment would not retroactively make this one successful. Any change to contexts, loss normalization, expert membership, shrinkage, bandwidth or the comparison collection must be named as a new configuration.

Quantile averaging and the experts' Gaussian approximations are further limits. An interval can achieve higher observed coverage by becoming too wide for useful planning. Evaluating an inventory action would require shortage and excess costs, decision lead times and prospective outcomes. Neither the generated examples nor real forecasting excerpts establish an operational benefit from this combination.

The latest related work also changes what an adequate next comparator would look like. A serial-error correction needs a completed previous combined error; a global feature learner needs an admissible reference corpus; a dynamic model average needs its own state and tuning policy. Giving one method information that another could not observe would test an information advantage rather than an allocation mechanism. The current study therefore preserves its measured gains while leaving these unexecuted comparisons unresolved. That boundary is preferable to calling a narrowly favorable table a completed comparison against the current forecasting literature.

10 Reproducibility and conclusions

Implementations and admitted numerical outputs are available in the frozen source and data snapshot.

The later paired-series analysis has its own retained numerical file and source identity. Its script and independent reproduction inputs are available in the paired-analysis source snapshot. The complete numerical archive retains each horizon explicitly. Its 10,584 weekly paths contain 137,592 scalar point predictions, and its 42,564 monthly paths contain 766,152. The distinct method-path and scalar counts describe computation volume; the paired analysis uses series as its resampling unit.

The recorded numerical environment uses NumPy 2.4.6, SciPy 1.15.3, StatsForecast 2.0.3 and Numba 0.68.1. The retained per-series worker durations have median and 90th percentile 3.815 and 6.485 seconds for weekly series, and 0.725 and 1.080 seconds for monthly series. The historical machine identity and end-to-end parallel makespan are unavailable. These per-worker durations do not recover wall time or support a speed comparison. The executable reconstruction and build commands, including explicit source, cache and output arguments, are in `next/REPRODUCING.md`. Numerical rescoring is distinct from a fresh expert fit.

The fixed rule is implemented in `data-pipeline/chronoscopelab/model/context_memory.py`; the temporally validated variants are in `data-pipeline/chronoscopelab/model/validated_combination.py`. Tests check complete historical eligibility, deterministic selection, convex constraints, quantile order and the fallback identity. Recursive expert loops are optionally compiled with Numba without fast-math transformations; independent reference comparisons

protect their forecast equations.

The external runner accepts explicit source, cache and output directories. Its cache binds source bytes, protocol, implementations and package versions. The full compressed numerical file is `manuscripts/context-memory/evidence/external-evaluation.json.gz`; it contains every outer forecast, target, scale, expert allocation, candidate validation score and temporal split. A separate checker reconstructs all source prefixes and independently rescores all 53,148 retained method-origin forecast paths and reconstructs weighted combinations from their stored expert paths. Browser documents retain eight decimal places for forecast paths; the full numerical file retains the original floating-point outputs. The selected-case comparison is reproduced separately from its original contiguous histories. The worked temporal-validation example executes the proposed variants on their declared prefixes.

The fixed contextual rule did not improve its recency-only ablation on the original selected blocks. ContextMemoryCV and ValidationMix have lower measured mean MASE than that control in both collections. ValidationMix expands the candidate set to include stacking and selects a stack at 289 of 882 weekly origins and 990 of 3,547 monthly origins. Relative to ContextMemoryCV, its mean MASE changes from .2848 to .2855 weekly and from .8996 to .8993 monthly. The incremental result is mixed. AutoTheta remains stronger on monthly point error and both collections' interval score. These outcomes support a qualified improvement in the evaluated combinations, with explicit remaining limits on component attribution, uncertainty and generalization.

A Fixed expert specifications

For every expert, quantile column q at lead j is $p_j + \Phi^{-1}(q)\hat{\sigma}\sqrt{j}$, at $q \in \{.1, .5, .9\}$. The shared Gaussian residual approximation is not a conformal or mixture interval. Population residual standard deviations use divisor equal to the number of retained residuals.

SeasonalNaive repeats the final m observations starting at the next observation's seasonal phase, with residuals $y_t - y_{t-m}$. SES initializes its level at the first observation and selects $a \in \{.05, .10, \ldots, .95\}$ by squared one-step error after that observation. Holt initializes level y_1 and trend $y_2 - y_1$; it uses additive level/trend updates, $a \in \{.1, .2, \ldots, .9\}$ and six equally spaced trend coefficients from .05 to .5. Its objective and residual dispersion exclude the first fitted value.

Additive HoltWinters initializes level at the first cycle mean, trend at the difference of the first two cycle means divided by m, and seasonal offsets from the first cycle. Its level grid is $\{.1, .3, .5, .8\}$, trend grid $\{.05, .2\}$ and seasonal grid $\{.1, .3, .6\}$. Squared errors and residual dispersion begin after the first cycle. At lead j, the seasonal index is $(n + j - 1) \bmod m$, including an incomplete terminal cycle. For $m < 2$ or fewer than two cycles it falls back to Holt.

The nonseasonal Theta expert uses the SES coefficient and level, OLS slope b and the finite-history equation [3]

$$p_j = \ell_n + \tfrac{1}{2} b \left[j - 1 + \frac{1 - (1 - a)^n}{a} \right]. \tag{25}$$

Its residual dispersion is the SES residual dispersion. This is distinct from the smoothing-plus-half-drift replay adaptation used in the separate streaming study. The source implementation fixes update ordering and first-minimum tie breaking in all grids; the same experts are used for every combination and ablation.

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